

PROJECT 2

EXPLORATORY DATA ANALYTICS (EDA) REPORT

The Online Retail Analysis was conducted using Exploratory Data Analysis (EDA) techniques in Excel to uncover trends, identify anomalies, and generate business insights.

STEPS TAKEN

1. DESCRIPTIVE STATISTICS OVERVIEW

- a. Numerical Columns were extracted from table and placed in a new sheet.
- b. Using basic Excel Function

In summary, the following were found:

- **Quantity:** Average purchase quantity is 2.95 units, most occurring number is 1, max 5, min 1, median 5.
- **Unit Price:** Mean price is approximately \$356, but highly variable. Prices range widely from \$11 to \$700, indicating diverse product categories.
- **Items in Cart:** Has an average of 5.5 items, with most customers buying between 1–10 items.
- **Total Price:**
Mean Revenue = \$1,054.
Some transactions are extreme outliers, with totals of up to \$3,456.

2. PRODUCT SALES INSIGHTS AND CONDITIONAL FORMATTING

Pivot Tables were created from cleaned datasets.

Insert Tab – Pivot Table – From Table/Range ▪ Insights were summarized using these Pivot Tables

Product Insight

- Top sellers by revenue: Chairs (\$195,620), Printers (\$195,613), and Laptops (\$192,127).
- Lowest revenue: Phones (\$151,722).
- Sales are relatively balanced across categories, but high-ticket items (Laptop, Printer) drive revenue concentration.

Payment Method Distribution

- **Most used:** Credit Card (234 orders, \$263,848 revenue) and Online (258 orders, \$262,443 revenue), although Cash approached Credit Card, getting more orders (246 orders), revenue was underperformed in comparison.
- **Least used:** Debit Card (\$232,361).
- Payment channels are evenly distributed, suggesting no strong customer preference bias.

Order Status

- Orders are spread across statuses:

Cancelled – 250

Returned – 247

Pending – 237

Shipped – 235

Delivered – 231

- High cancellation/return rates may indicate fulfillment or product satisfaction issues.

Marketing Effectiveness

- Top channel: Instagram (\$275,285 revenue, 259 orders).
- Strong performers: Email (\$261,809) and Google (\$250,441).
- Weakest: Referral (\$226,816).
- Social media (Instagram, Facebook) drives significant sales, but referrals underperform.

3. OUTLIER ANALYSIS

- Quartile bounds:

Q1 = \$410.52

Q3 = \$1,578.48

IQR = \$1,167.96

- Upper bound for normal transactions: \$3,330.41
- Transactions above this, for example, \$3,456, are flagged as outliers.
- These outliers suggest promotional stacking or bulk buying behavior.